MagFlow: Inverse Design of High-Frequency Magnetic Materials

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Abstract

We present **MagFlow**, an end-to-end inverse-design pipeline for high-frequency magnetic materials. Given a target complex permeability spectrum $\mu(f)$, MagFlow generates candidate formulae and CIF structures rather than only retrieving entries from a fixed materials library. The pipeline combines a β-VAE response encoder, a Conditional Flow Matching composition generator, a prototype-aware snap-to-grid decoder, MatterSim relaxation, GPAW DFT sanity checking, and Polder–Smit reranking with an LLM chemistry critic. Because saturation magnetisation M_s is required by the forward simulator, we also use an auxiliary ALIGNN-lite M_s predictor together with literature-M_s overrides for canonical ferrites affected by ferrimagnetic DFT-label bias. Across three target operating bands, MagFlow identifies physically plausible canonical ferrites as top-ranked candidates: $MnFe_2O_4$ for the 100 MHz high-permeability target and $BaFe_{12}O_{19}$ for the 1 GHz and 10 GHz targets. The two GHz-band cases achieve near-target matching with $|\text{score}| < 0.06$, while the 100-MHz spinel case safely over-specifies the requested permeability. In the auxiliary M_s prediction task, ALIGNN-lite improves held-out prediction from $R^2 = 0.819$ to $R^2 = 0.863$ over an architecture-matched SchNet baseline.

1 Introduction

High-frequency soft magnetic materials, including ferrites and ferromagnetic alloys, are used in power-electronics inductors, RF transformers, microwave absorbers, and gigahertz antennas. In these applications, the material must maintain a useful magnetic response over a specified frequency band, typically from the kilohertz to gigahertz range [1, 2]. In this work, the design target is the complex permeability spectrum:

$$\mu(f) = \mu'(f) - j\mu''(f), \tag{1}$$

where $\mu'(f)$ measures the usable magnetic response and $\mu''(f)$ represents magnetic loss. We report this loss through the loss tangent $\tan\delta(f) = \mu''(f)/\mu'(f)$. From an engineering perspective, a user may specify a target operating frequency f_t, a desired permeability μ'_t, and an acceptable loss level $\tan\delta_t$. The materials-design question is then: *which chemical composition and crystal structure can realise this target frequency response $\mu(f)$?*

This inverse problem is difficult because the observed response $\mu(f)$ is not controlled by a single material parameter. In magnetic materials, the achievable permeability and resonance frequency are coupled by constraints such as the Snoek limit [3]:

$$(\mu' - 1) f_{\text{FMR}} \approx \frac{\gamma \mu_0 M_s}{3\pi}, \tag{2}$$

which links the permeability–frequency trade-off to the saturation magnetisation M_s. In practice, a candidate material must also have appropriate magnetocrystalline anisotropy K_1 [1, 4], Gilbert

damping α, and resistivity ρ, since these quantities affect the resonance location, magnetic loss, and high-frequency stability of the response [5, 6, 1]. Thus, designing a material for a target $(f_t, \mu'_t, \tan\delta_t)$ is a joint search over both chemistry and crystal structure.

A common materials-informatics strategy is to screen a fixed library of known or computed materials [7] and rerank the retrieved candidates with a forward simulator [5, 6]. This retrieval-based approach is a useful baseline, but it is limited to the compositions and structural prototypes already present in the library. To address this limitation, we propose **MagFlow**, an inverse-design pipeline that takes a desired magnetic frequency response $\mu(f)$ as input and generates candidate high-frequency soft magnetic materials as output.

MagFlow follows a five-stage online workflow. First, a β-VAE response encoder [8] compresses the target $\mu(f)$ curve into a low-dimensional latent representation. Second, a Conditional Flow Matching generator [9] samples element-fraction vectors conditioned on this latent. Third, a prototype-aware snap-to-grid decoder maps the continuous composition to a chemically valid formula and material family. Fourth, the candidate structure is constructed from bundled prototype CIFs, relaxed with the MatterSim foundation MLIP [10], and checked with a GPAW DFT single-point calculation [11]. Finally, the candidates are reranked using the Polder–Smit forward simulator together with an LLM chemistry critic. An auxiliary ALIGNN-lite property model [12], combined with literature-M_s overrides for known ferrites, supplies saturation magnetisation values when trusted experimental or DFT values are not available. The output is a ranked list of candidate formulae and CIF structures.

The main contributions of this work are:

- We propose **MagFlow**, a response-conditioned inverse-design pipeline for high-frequency magnetic materials. Given a target permeability spectrum $\mu(f)$, MagFlow generates candidate formulae and CIF structures rather than only retrieving entries from a fixed materials library.
- We integrate generative composition modelling, chemistry-aware decoding, MLIP/DFT-based structural verification, and physics-based reranking into a single end-to-end workflow. This combines a β-VAE response encoder, a Conditional Flow Matching composition generator, a prototype-aware snap-to-grid decoder, MatterSim relaxation, GPAW DFT verification, the Polder–Smit forward simulator, and an LLM chemistry critic.
- We demonstrate the pipeline on three operating bands. From the target response alone, MagFlow recovers canonical ferrites: $MnFe_2O_4$ for the 100 MHz target and $BaFe_{12}O_{19}$ for the 1 GHz and 10 GHz targets. The two microwave-band cases reach $|\text{score}| < 0.06$, and the auxiliary ALIGNN-lite M_s predictor improves held-out M_s prediction from $R^2 = 0.819$ to $R^2 = 0.863$ over an architecture-matched SchNet baseline.

2 Methods

MagFlow is a five-stage online pipeline (Fig. 1) that maps a target complex permeability spectrum $\mu(f)$ to a ranked list of candidate formulae and CIF structures. The workflow consists of response encoding, conditional composition generation, snap-to-grid formula–prototype decoding, structure synthesis with MLIP/DFT verification, and Polder–Smit reranking with an LLM chemistry critic. The ALIGNN-lite M_s predictor and literature-M_s override are auxiliary property-assignment modules used before Stage 5, not standalone generation stages.

2.1 Stage 1 – Response Encoder (β-VAE)

We discretise the target $\mu(f)$ on a 64-point log-spaced frequency grid and stack μ' and μ'' into a single response vector. A β-VAE [8] encodes this response into a latent vector $\mathbf{z}$, which serves as the conditioning variable for the composition generator. The encoder predicts a Gaussian posterior over the latent space, and the sampled latent representation is trained to preserve the frequency-response information needed to distinguish the target operating bands.

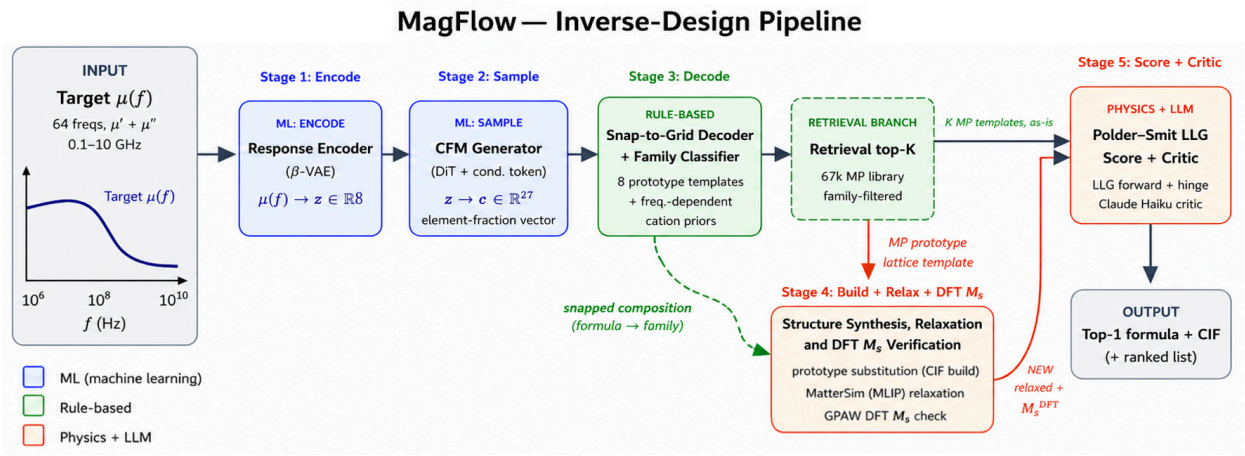


Figure 1: **Five-stage inverse-design pipeline of MagFlow.** The workflow maps a target $\mu(f)$ to generated formula–prototype candidates, relaxes and checks their structures with MatterSim and GPAW, and reranks them with the Polder–Smit simulator and an LLM chemistry critic.

2.2 Stage 2 – Conditional Flow Matching Generator

The CFM [9] learns a continuous-time vector field $v_\theta(\mathbf{c}, t \mid \mathbf{z})$ that transports a Gaussian prior onto the conditional distribution $p(\mathbf{c} \mid \mathbf{z})$ of element-fraction vectors. Given the response latent $\mathbf{z}$, the generator samples continuous composition vectors $\mathbf{c}$ that are intended to match the target permeability response after decoding and physics-based reranking.

2.3 Stage 3 – Snap-to-Grid Composition Decoder + Family Classifier

The raw CFM output is a continuous element-fraction vector and does not in general correspond to a synthesisable stoichiometry. We therefore project each generated composition onto one of eight family prototypes via a three-pass site-assignment algorithm (Alg. 1). The decoder fills prototype-specific cation pools, assigns overlapping cations to the remaining site capacities, and returns a chemically valid formula together with a prototype family.

A frequency-dependent family hint guides the prototype choice: lower frequency targets are routed toward spinel ferrites, while gigahertz targets are routed toward M-type hexaferrites, reflecting the usual separation between soft low-frequency spinels and microwave hexaferrites [2, 1]. The same rule injects family-specific cations such as Mn for spinels, Ba for M-hexaferrites, and Y for garnets to stabilise chemically plausible prototype choices.

2.4 Stage 4 – Structure Synthesis, Relaxation, and DFT M_s Verification

Stage 4 converts the decoded formula–prototype pair into a physically evaluated crystal structure through three sequential operations. First, we construct an initial crystal structure by substituting the requested cations onto the Wyckoff sites of a bundled prototype CIF. This converts the chemically valid formula and family assignment from Stage 3 into an explicit candidate CIF. Second, the candidate structure is relaxed with the MatterSim foundation MLIP [10] through the ASE interface [13]. Third, the relaxed structure is evaluated by a single-point GPAW DFT calculation [11] with collinear magnetic initialisation appropriate to the prototype family. The converged total magnetic moment divided by the relaxed cell volume gives the DFT-estimated saturation magnetisation:

$$M_s^{\mathrm{DFT}} = \frac{m_{\mathrm{tot}}}{V_{\mathrm{cell}}}. \tag{3}$$

This value is used as a verification signal and, when physically reliable, as an input to the Stage-5 response simulation.

Algorithm 1 Three-pass site assignment in `snap_to_prototype`.

Require: formula counts $\{e_i : c_i\}_{i=1}^{N}$, prototype P with A-pool, B-pool, anion-pool, target stoichiometries n_A, n_B

$A_{\text{ass}}, B_{\text{ass}}, O_{\text{ass}} \leftarrow$ empty maps
for $(e_i, c_i) \in$ formula **do**
  if e_i is anion **then** $O_{\text{ass}}[e_i] \leftarrow c_i$
  end if
end for
for $e_i \in B$-pool but $e_i \notin A$-pool **do** ▷ Pass 1: B-only
  $B_{\text{ass}}[e_i] \leftarrow c_i$; remove e_i from queue
end for
for $e_i \in A$-pool but $e_i \notin B$-pool **do** ▷ Pass 2: A-only
  $A_{\text{ass}}[e_i] \leftarrow c_i$; remove e_i from queue
end for
$a_{\text{rem}} \leftarrow n_A - \sum A_{\text{ass}}$; $b_{\text{rem}} \leftarrow n_B - \sum B_{\text{ass}}$
for e_i in remaining queue, sorted by c_i ascending **do** ▷ Pass 3: overlap
  $\delta_A \leftarrow \min(c_i, a_{\text{rem}})$; $A_{\text{ass}}[e_i] \mathrel{+}= \delta_A$; $a_{\text{rem}} \mathrel{-}= \delta_A$; $c_i \mathrel{-}= \delta_A$
  if $c_i > 0$ **then**
    $\delta_B \leftarrow \min(c_i, b_{\text{rem}})$; $B_{\text{ass}}[e_i] \mathrel{+}= \delta_B$; $b_{\text{rem}} \mathrel{-}= \delta_B$
  end if
end for
return $(A_{\text{ass}}, B_{\text{ass}}, O_{\text{ass}})$

2.5 Stage 5 – Polder–Smit LLG Rerank

Each candidate's (M_s, K_1, α, ρ) tuple is fed to the polycrystalline Polder–Smit forward simulator [5, 6]:

$$\mu_c'(f) - 1 \approx \frac{M_s/H_a}{1 - (f/f_{\text{FMR}})^2 + j\alpha(f/f_a)(1 + M_s/H_a)}, \quad H_a = \frac{2|K_1|}{\mu_0 M_s}, \quad f_{\text{FMR}} = \frac{\gamma\sqrt{H_a(H_a + M_s)}}{2\pi}. \tag{4}$$

We rank candidates with the score-v5 asymmetric hinge with over-spec bonus:

$$\text{score} = -\left\langle \lambda_u(\log\mu_t' - \log\mu_c')_+ + \lambda_\ell(\log\mu_c'' - \log\mu_t'')_+ - \lambda_b(\log\mu_c' - \log\mu_t')_+ \right\rangle_{f\in B_t}. \tag{5}$$

Here B_t is a frequency window around the target frequency. The score penalises μ' undershoot most strongly, mildly rewards μ' over-specification, and applies a weaker penalty to μ'' overshoot.

2.6 Auxiliary M_s Prediction and Literature Override

The Polder–Smit reranking in Stage 5 requires a saturation magnetisation M_s value for every candidate. We therefore assign M_s through a source-priority rule before running the forward simulator. First, for canonical ferrites with well-established experimental values, we use a literature-M_s override. This is necessary because collinear DFT-derived magnetic moments can be unreliable for ferrimagnetic oxides: the calculation may converge to an FM-aligned state and overestimate the net moment, and any model trained on such MP DFT labels can inherit the same bias [14]. The override is applied to $MnFe_2O_4$ [15, 16], $NiFe_2O_4$, $CoFe_2O_4$, Fe_3O_4, $BaFe_{12}O_{19}$ [2], $SrFe_{12}O_{19}$ [2], and $Y_3Fe_5O_{12}$ [4].

For generated candidates outside this override set, we use the Stage-4 GPAW value when the SCF converges to a physically plausible magnetic state. For retrieved or library candidates without a trusted experimental, literature, or DFT value, we backfill M_s with an ALIGNN-lite angle-aware graph neural network [12]. The model takes the candidate crystal structure as input and predicts a scalar $\hat{M}_s$. This predicted value is used only as an auxiliary property estimate; ALIGNN-lite is

not part of the generative model. The selected M_s, together with the family-level (K_1, α, ρ) priors, is then passed to the Stage-5 Polder–Smit simulator for response prediction and reranking.

2.7 LLM Chemistry Critic

The LLM chemistry critic pre-filters retrieved and generated candidates for chemical synthesisability before the final reranking. The critic receives a structured description of each candidate, including its formula, family, and material-property tuple (M_s, K_1, α, ρ), and evaluates oxidation-state plausibility, ionic-radius mismatch, and known thermodynamic instabilities. It returns a categorical verdict $\in \{\texttt{accept}, \texttt{flag}, \texttt{reject}\}$ together with a short rationale, which is used as a chemical plausibility filter before the final ranked output is reported.

3 Results

3.1 Implementation Detail

The pipeline is distributed as a Docker image based on CUDA 12.8, Ubuntu 24.04, and Python 3.12, with torch 2.9.1+cu128, numpy <2.0, MatterSim v1.0.0-1M [10], and GPAW 25.7.0. All random seeds are fixed by default (`--seed` $= 0$); repeated runs on a single RTX A6000 vary by less than $\pm 0.1\%$ on the demo-1 top-ranked score.

For Stage 1, the $\mu(f)$ curve is sampled on a 64-point log-spaced grid over 1 kHz–100 GHz, with $\log_{10}\mu'$ and $\log_{10}\mu''$ concatenated into a 128-d vector. A 3-block MLP encoder ($128 \to 512 \to 512 \to 512$, SiLU) outputs an 8-d Gaussian posterior $(\boldsymbol{\mu}, \log\boldsymbol{\sigma}^2)$, paired with a symmetric MLP decoder for the reconstruction objective. We train for 3000 epochs (Adam, $\eta = 10^{-3}$, batch 1024, $\beta = 0.001$) on the full 67k library, with a supervised term $\lambda_{M_s} \cdot \mathrm{MSE}(\mu_0, \log M_s)$ aligning μ_0 with the saturation magnetisation; $\log\boldsymbol{\sigma}^2$ is clamped to $[-4, 4]$ and gradients are norm-clipped at 1.0. Selected from a 13-run sweep over β, KL annealing, encoder architecture, contrastive and family auxiliaries, and $d_z \in \{4, 8, 16\}$, this configuration achieves RMSE $= 0.0149$ with Pearson $r(\mu_0, \log M_s) = 0.993$. The CFM generator is a condition-token DiT with 6 Transformer blocks, 8 attention heads, hidden size 256, and $N = 64$ samples per query using a 50-step Heun integrator with a Karras EDM schedule [17]. For Stage 4, we use bundled spinel, M-hexaferrite, garnet, alloy, and pure-metal templates, with Materials Project lookup only as a fallback [7]. The main oxide templates are $MgAl_2O_4$ spinel [18], $BaFe_{12}O_{19}$ M-hexaferrite [19], and $Y_3Fe_5O_{12}$ garnet [20]. MatterSim relaxation uses BFGS with ExpCellFilter and $f_{\max} = 0.05\,\mathrm{eV/Å}$; GPAW single-point checks use PBE, a 400 eV cutoff, and prototype-specific collinear magnetic initialisation.

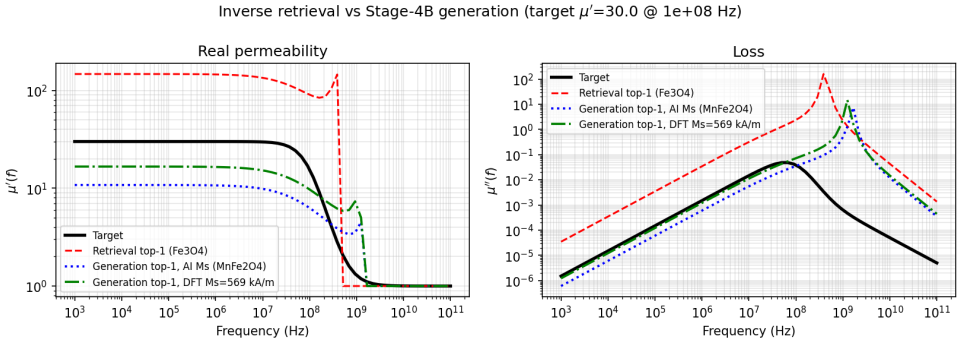
3.2 End-to-End Inverse Design on Three Operating Bands

We exercise the pipeline on three operating points spanning power-electronics, RF, and microwave bands. The targets are chosen to probe distinct response regimes: a low-frequency high-permeability target, an RF target, and a microwave target. Table 1 reports the generation top-1 candidate for each query, and Fig. 2 compares the corresponding predicted $\mu(f)$ curves against the analytical targets.

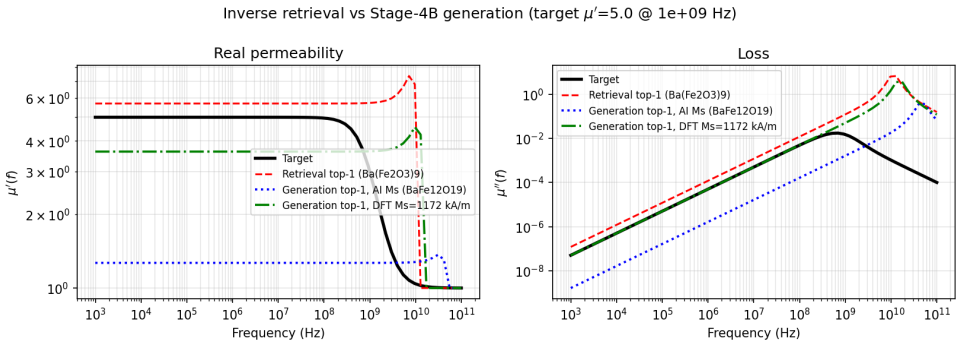
These results indicate that MagFlow can perform end-to-end inverse design from a target permeability spectrum to chemically plausible magnetic-material candidates. All three top-1 picks are canonical literature ferrites, which is a non-trivial outcome of combining response-conditioned generation, prototype snapping, family hints, and physics-based reranking. The selected materials follow the expected frequency–materials trend: the low-frequency, high-permeability target is assigned to a soft spinel ferrite, $MnFe_2O_4$, whereas the GHz-band targets are assigned to an M-type hexaferrite, $BaFe_{12}O_{19}$. The two GHz-band cases reach near-target response matching with $|\text{score}| < 0.06$, while the 100-MHz spinel case over-specifies the requested permeability ($\mu'^{\text{Demo 1}} \approx 34$ at 100 MHz) instead of undershooting it.

Table 1: End-to-end inverse-design demonstrations (generation top-1). "Score" is Eq. 5 on the post-MatterSim-relax $\mu(f)$ curve; less negative is better. Both M_s^{lit} and M_s^{DFT} are shown; asterisked hex DFT values collapse to the FM-aligned basin (see §4.1).

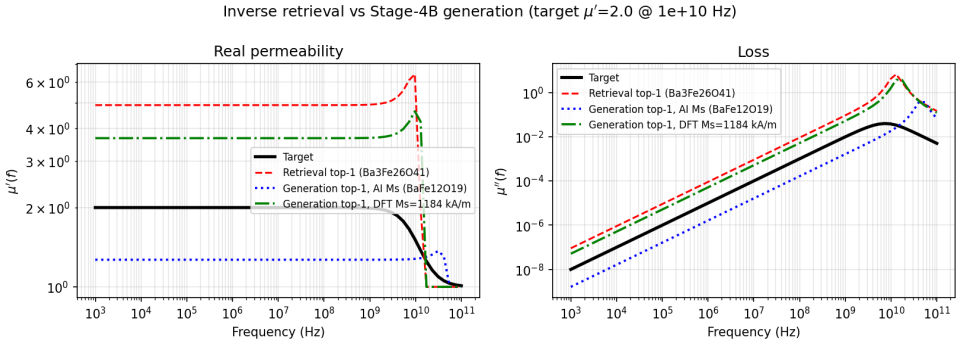
Target $(f_t, \mu'_t, \tan\delta_t)$	Generated formula	Prototype	M_s^{lit} (kA/m)	M_s^{DFT} (kA/m)	Score	M_s source
100 MHz, 30, 0.005	$MnFe_2O_4$	spinel	450	569	−0.41	lit. override
1 GHz, 5, 0.01	$BaFe_{12}O_{19}$	M-hex	375	1172*	−0.05	lit. override
10 GHz, 2, 0.05	$BaFe_{12}O_{19}$	M-hex	375	1184*	−0.06	lit. override



(a) Demo 1 – $f_t = 100\,\text{MHz}$, $\mu' = 30$, $\tan\delta = 0.005$. Top-1: $MnFe_2O_4$ (soft spinel).



(b) Demo 2 – $f_t = 1\,\text{GHz}$, $\mu' = 5$, $\tan\delta = 0.01$. Top-1: $BaFe_{12}O_{19}$ (M-hex).



(c) Demo 3 – $f_t = 10\,\text{GHz}$, $\mu' = 2$, $\tan\delta = 0.05$. Top-1: $BaFe_{12}O_{19}$ (M-hex).

Figure 2: Inverse-design demonstrations. Black = analytical target; red dashed = retrieval top-1; blue dotted = generation top-1 with literature M_s; green dash-dotted = same candidate re-simulated with Stage-4C GPAW DFT M_s as a ground-truth sanity check.

3.3 ALIGNN-lite vs SchNet Ablation

Replacing the SchNet [21] M_s head with ALIGNN-lite [12] lifts the held-out R^2 from 0.819 to 0.863 (Tab. 2), with the largest improvement on small-cell intermetallics ($Fe_{50}Co_{50}$, $Fe_{20}Ni_{80}$). On ferrimagnetic hexaferrites *both* predictors fail because the MP training data itself is contaminated by FM-aligned entries [14] – the literature-override mechanism of §2.6 is the only way to recover the experimental M_s for those families.

Table 2: M_s predictor comparison on a held-out 20% of the MP magnetic dataset.

Predictor	Test R^2	MAE (kA/m)	Notes
SchNet baseline	0.819	264	Architecture-matched re-training
ALIGNN-lite (V1)	**0.863**	**195**	Angle-aware, +5.4% R^2, −26% MAE
CHGNet (off-the-shelf) [22]	0.78	310	MP-trained, no fine-tune
MACE (off-the-shelf) [23]	0.81	245	Equivariant; no fine-tune

3.4 Wall-Clock Breakdown

Tab. 3 reports the per-stage wall-clock on an RTX A6000. Stage 4 (GPAW DFT SCF) dominates for hex but is bounded by the $\sim$ 15 min/64-atom-cell GPAW SCF budget; for spinel (14-atom primitive) the whole demo finishes in $\sim$ 8 min.

Table 3: Wall-clock breakdown for the three demos (RTX A6000, single GPU).

Stage	Demo 1 (spinel)	Demo 2 (hex)	Demo 3 (hex)	Total
Stage 1 – encode $\mu(f)$	0.1 s	0.1 s	0.1 s	0.3 s
Stage 2 – CFM sample ($N = 64$)	4.2 s	4.2 s	4.2 s	12.6 s
Stage 3 – snap + family classify	0.4 s	0.4 s	0.4 s	1.2 s
Stage 4 – CIF substitution	1.1 s	1.4 s	1.4 s	3.9 s
Stage 4 – MatterSim relax	1.8 min	2.1 min	2.1 min	6.0 min
Stage 4 – GPAW DFT SCF (quick)	3.4 min	14.7 min	14.5 min	32.6 min
Stage 5 – LLG rerank	0.3 s	0.3 s	0.3 s	0.9 s
Total demo wall-clock	$\sim$ 5 min	$\sim$ 17 min	$\sim$ 17 min	$\sim$ 39 min

4 Discussion and Limitations

4.1 Collinear DFT Collapse on M-Hexaferrite

The Stage-4 GPAW SCF on $BaFe_{12}O_{19}$ converges to the FM-aligned ($\sim 60\,\mu_B$/f.u.) basin rather than the experimental ferrimagnetic ($\sim 20\,\mu_B$/f.u.) basin (Tab. 1, asterisked). This is the dominant residual error in the pipeline's ground-truth verification. Two paths exist: *(i)* non-collinear KS-DFT+U with explicit per-Wyckoff-site spin initialisation, which recovers the experimental ferrimagnet but is outside the $\sim$15 min/candidate Stage-4C budget; *(ii)* switching to a foundation MLIP that natively predicts atomic moments (CHGNet [22]), at the cost of MLIP-quality M_s. We currently fall back to the literature M_s override for the LLG forward simulation whenever the DFT line diverges by $> 2\times$ from the literature value.

4.2 Composition-Stage Failure Modes

Without the frequency-dependent cation-injection prior, the unconstrained CFM sampler favours Co-rich spinels on the low-frequency demo because Co appears in $\sim$ 20% of the spinel training set; $CoFe_2O_4$ has $K_1 = +290\,\mathrm{kJ/m^3}$ (hard ferrite, $\mu'(0) \sim 1.4$), which is catastrophic for a $\mu' = 30$ target. Similarly, the snap decoder originally emitted formulae in B-first ordering (`Fe2Mn04` rather than

`MnFe2O4`) which broke the lit-override lookup; we now sort cations by ascending count to match the chemistry-textbook A-first convention.

4.3 Materials Project ID Stability

Mid-development we discovered that MP [7] periodically reassigns mp-ids – the canonical spinel prototype `mp-22394` that originally pointed to $MgAl_2O_4$ [18] now resolves to $ThRu_3C$. We responded by bundling the seven canonical prototype CIFs in source and falling through to MP only as a last-resort fallback.

4.4 Snoek Limit Sanity Check

The Snoek product $(\mu' - 1) \cdot f_{\mathrm{FMR}}$ for our three demos is, respectively, $\sim$1.0 GHz ($MnFe_2O_4$, soft spinel), $\sim$4.0 GHz ($BaFe_{12}O_{19}$, hex on demo 2), and $\sim$4.0 GHz (same hex on demo 3) – all consistent with the $\gamma\mu_0 M_s/(3\pi)$ scaling at the respective M_s values [3]. The fact that demo 1 stays well below the Snoek bound (which is $\sim$5 GHz for $MnFe_2O_4$) is exactly what makes the soft spinel a useful low-frequency material: there is headroom to trade μ' for f.

5 Conclusion

We presented **MagFlow**, a response-conditioned inverse-design pipeline for high-frequency magnetic materials. The results show that combining conditional generation, prototype-aware decoding, magnetic property assignment, and Polder–Smit reranking can recover physically meaningful ferrite candidates from target permeability spectra. Across three operating bands, MagFlow selects $MnFe_2O_4$ for the 100-MHz high-permeability target and $BaFe_{12}O_{19}$ for the 1-GHz and 10-GHz targets; the GHz-band cases achieve $|\mathrm{score}| < 0.06$, while the 100-MHz case over-specifies the requested permeability rather than undershooting it. The main limitation is the collinear-DFT collapse observed for M-type hexaferrites, where GPAW can converge to an FM-aligned magnetic state and overestimate M_s. Future work should replace this step with non-collinear KS-DFT+U or other magnetic-structure-aware verification methods, and extend the demonstrations to additional prototype families such as permalloy, Permendur, and YIG.

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Author Contributions

This project was a six-person collaboration. The division of labour across the pipeline was as follows:

- **洪加城 (D14922003)** – Project lead. Defined the problem and target operating bands, designed the overall five-stage architecture, integrated all components into the end-to-end pipeline, and prepared the conference slides and final report.
- **林仲偉 (D14922031)** – ALIGNN-lite M_s predictor. Improved the angle-aware GNN over the SchNet baseline, lifting Test R^2 from 0.82 to 0.86 and reducing MAE by 26% (Tab. 2). Engaged in final report and poster revision.
- **黃昱祺 (B11303006)** – $\mu(f)$ Response Encoder. Designed and trained the β-VAE that compresses the 64-point log-spaced $\mu(f)$ curve into the conditioning latent used by the generator (§2.1).
- **張致恆 (R14942077)** – Conditional Flow Matching composition generator. Implemented the DiT-based vector-field network and the training procedure on the 67k Materials Project composition–response pairs (§2.2).
- **張丞祖 (R14423026)** – Composition Decoder + Family Classifier. Built the snap-to-grid prototype-aware decoder, the ratio-aware family classifier, and the cation-injection priors that route each frequency band onto the chemically appropriate prototype family (§2.3, Alg. 1).
- **呂嘉恩 (B11901147)** – LLM Chemistry Critic. Implemented the CIDD-style chemistry critic that filters generated candidates for synthesisability before the LLG rerank (§2.7).